Answers & Solutions:

1. **A. electrons**

2. **C. the EB junction must be forward-biased and the CB junction must be-biased.**

3. **B. a stable Q point**

4. **A. 1.25 A**

Solution:

$$I_E = I_B + I_C$$
$$I_E = 50\text{ mA} + 1.2\text{ A}$$
$$I_E = 1.25\text{ A} \rightarrow \text{Ans.}$$

5. **D. the arrow points in on the emitter lead.**

6. **B. I_E.**

7. **A. 361.6 kΩ ; 2.4 kΩ**

Solution:

$$I_B = \frac{I_C}{\beta} = \frac{2.5\text{ mA}}{80} = 31.25\ \mu\text{A}$$
$$R_B = \frac{V_{RB}}{I_B} = \frac{V_{CC} - V_{BE}}{I_B} = \frac{12\text{ V} - 0.7\text{ V}}{31.25\ \mu\text{A}}$$
$$= 361.6\text{ k}\Omega \rightarrow \text{Ans.}$$
$$R_C = \frac{V_{RC}}{I_C} = \frac{V_{CC} - V_C}{I_C} = \frac{V_{CC} - V_{CEQ}}{I_{CQ}}$$
$$= \frac{12\text{ V} - 6\text{ V}}{2.5\text{ mA}} = 2.4\text{ k}\Omega \rightarrow \text{Ans.}$$

8. **D. collector-base**

9. **D. 1.52 mA**

Solution:

$$I_E = \frac{I_C - I_{CEO}}{\alpha}$$
$$I_E = \frac{1.5\text{ mA} - 1\ \mu\text{A}}{0.984}$$
$$I_E = 1.52\text{ mA} \rightarrow \text{Ans.}$$
where:
$$\alpha = \frac{\beta}{1+\beta} = \frac{60}{1+60}$$
$$\alpha = 0.984$$

10. **D. hold the circuit stable at the designed Q-point**

11. **B. 4 mW**

Solution:

$$P_{D(ave)} = \left(\frac{t_{on}}{T}\right) V_{CE(sat)} I_{C(sat)}$$
$$= \left(\frac{1\mu s}{5\mu s}\right)(0.2\text{ V})(100\text{ mA})$$
$$P_{D(ave)} = 4\text{ mW} \rightarrow \text{Ans}$$

12. **C. I_C and I_E.**

13. **A. saturation and cutoff**

14. **A. the common-emitter amplifier.**

15. **A. the emitter region.**

16. **A. 2.78 mA**

Solution:

$$I_E = I_C + I_B$$
$$I_C = I_E - I_B$$
$$I_C = 2.8\text{ mA} - 0.02\text{ mA}$$
$$I_C = 2.78\text{ mA} \rightarrow \text{Ans.}$$

17. **C. 0.7 V across each depletion layer**

18. **D. the base region.**

19. **D. Both A and C.**

20. **D. base bias.**

21. **C. V_{CE} = 0 V.**

22. **B. V_{CE} = V_{CC}.**

23. **B. 3.992 mA**

Solution:

$$I_C = \alpha I_E = (0.998)(4\text{ mA})$$
$$I_C = 3.992\text{ mA} \rightarrow \text{Ans.}$$

24. **A. crossover distortion**

25. **B. The emitter follower**

26. **A. stabilization**

27. **A. 3.25 mA**

Solution:

$$\alpha = \frac{I_{CQ}}{I_{EQ}}$$
$$I_{EQ} = \frac{I_{CQ}}{\alpha} = \frac{3.12\text{mA}}{0.96}$$
$$= 3.25\text{ mA} \rightarrow \text{Ans.}$$
where:
$$I_{CQ} = \beta I_{BQ(act.)} = (24)(130\text{x}10^{-6})$$
$$I_{CQ} = 3.12\text{ mA}$$
$$I_{BQ(act)} = I_{BQ} + I_{CBQ} = 120\ \mu\text{A} + 10\ \mu\text{A}$$
$$I_{BQ(act)} = 130\ \mu\text{A}$$
$$\beta = \frac{\alpha}{1-\alpha} = \frac{0.96}{1-0.96} = 24$$

28. **D. its low input impedance.**

29. **B. collector is positive with respect to base**

30. **A. the active region.**

31. **B. the collector region.**

32. **C. The common-emitter amplifier**

33. **A. A_v decreases.**

34. **A. The common-base amplifier**

35. **B. the common-collector amplifier.**

36. **B. Active region**

37. **B. $I_{C(Sat)}$ and $V_{CE(off)}$.**

38. **A. Forward bias the base/emitter junction and reverse bias the base/collector junction.**

39. **A. I_C/L_E.**

40. **A. The common-base amplifier**

41. **C. V_c decreases.**

EXCEL REVIEW CENTER PHILIPPINES

42. **A. cut off.**
43. **D. Both A and B.**
44. **C. low resistance**
45. **D. two p-n junctions.**
46. **B. very stable Q point.**
47. **D. Both A and B.**
48. **C. emitter current**
49. **A. the common-collector amplifier.**
50. **C. collector current and collector to emitter voltage**

END